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MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Physics

End Term Examination (Even Semester / 2025-2026)

Course: Modern Physics (B.Tech.)
Code: 22PHT102

Max. Marks: 50
Time: 2 hr. 30 min

Answer all questions.

- Q1. (a) Show that a particle confined in an infinite one-dimensional well has discrete energy levels. Find the normalized wave function. [6]
(b) Prove that the group velocity of the wave packet corresponding to a particle is equal to the particle's velocity (For both relativistic and non-relativistic cases). [4]
- Q2. (a) Discuss any two applications of Heisenberg's uncertainty principle. [6]
(b) Explain the concept of quantum tunnelling and orthogonality of wave functions. [4]
- Q3. (a) An electron is constrained by impenetrable walls to move in a rectangular box of sides $1\text{\AA}$, $2\text{\AA}$ and $3\text{\AA}$. The potential energy of the particle inside the box is zero. Calculate the energy of the ground state and the first excited state. Will there be any degenerate state? [3]
(b) Consider a one-dimensional particle that is confined within the region $0 \leq x \leq a$ and whose wave function is $\psi(x, t) = \sin\left(\frac{\pi x}{a}\right) e^{-i\omega t}$ [3+4]
(i) Find the potential $V(x)$
(ii) Calculate the probability of finding the particle in the interval $a/4 \leq x \leq 3a/4$.
- Q4. (a) State and prove the Poynting's theorem. [6]
(b) Show that the electromagnetic waves are transverse in nature. [4]
- Q5. (a) Calculate the de Broglie wavelength for a proton of kinetic energy 70 MeV, and a 100 g bullet moving at 900 m/s. Comment on the result. [3+1]
(b) The electric and magnetic fields of a specific configuration are $\mathbf{E} = A_0 \omega \cos(kx - \omega t) \hat{y}$ and $\mathbf{B} = A_0 k \cos(kx - \omega t) \hat{z}$. Show that they satisfy Maxwell's equations in free space (consider $\omega = ck$). [6]
- 6.61
9
276
91 662
63
3
2
- 10-68
10-314
19+4
- 144+9+4
144
144
- 10-68
10-314
19+4

Malaviya National Institute of Technology Jaipur
Department of Physics
End-term examination 2025-26

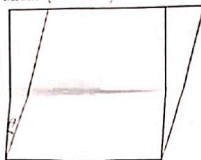
Class: B.Tech., 1st semester.
Course Title: Classical Physics.
Course Code: 22PHT101.

Date: November 26, 2025.
Time duration: 2 hour 30 minutes.
Maximum Marks: 50.

Note: Attempt all questions. There are 7 questions in total.

- ✓ Q.1) Obtain the expression for velocity and acceleration in plane polar coordinates. 5 marks
- Q.2) (a) An Atwood machine is placed on a weighing machine and its weight is noted. Then the masses are released and weight noted once again. Is the new reading shown on the machine different from the first one? Arrive at a conclusion using physical principles alone, and then verify your answer using calculations based on a free body diagram. You can make the standard assumptions to simplify your calculations. 3 marks
- (b) A rocket is lifted-off the surface of the earth with no horizontal motion at all. The exhaust velocity of the fuel expelled by the rocket is $\vec{u}$ (constant) relative to the rocket. The mass and the velocity of the rocket is M and $\vec{v}$ at time t respectively. Obtain the equation of motion of the rocket. If the initial velocity of the rocket at time $t = 0$ is zero and the initial mass is M_0 , find the final velocity in terms of the final mass at time $t = t_f$ assuming a constant gravitational field. 6 marks
- ✓ Q.3) A harmonic oscillator is driven by a force with amplitude F_0 and frequency ω . Obtain the equation of motion with no damping and find the solution. Explain how the amplitude of the harmonic oscillator varies with respect the frequency of the driving force. 7 marks
- Q.4) (a) Obtain the moment of inertia of a solid sphere of mass M and radius R about its diameter. 5 marks
- ✓ (b) A solid cylinder of mass 200 kg rotates about its axis with an angular speed of 100 m/s. If the radius of the cylinder is 0.5 m, calculate the angular momentum of the cylinder about its axis. 2 marks
- ✓ Q.5) (a) Obtain the expression for the angular frequency of precession of a gyroscope. 4 marks
- (b) A uniform disc of diameter 300 mm and of mass 5 kg is mounted on one end of an arm of length 600 mm. The other end of the arm is free to rotate in a universal bearing. If the disc rotates about the arm with a speed of 300 r.p.m. clockwise, looking from the front, with what speed will it precesses about the vertical axis? 3 marks
- ✓ Q.6) (a) Explain the behaviour of a wire under increasing stress with the help of stress-strain curve. 5 marks

- ✓(b) A shear force of magnitude 10 N is applied to a box, which was initially a cube of 50 cm side. The shear modulus of the material of the box is 10^{10} N/m². Find the value of shear ($\theta \times 10^{10}$).



2 marks

- Q.7) (a) Show that the force per unit volume on a fluid element is negative of the pressure gradient. Also show that the surface of a fluid rotating with an angular velocity Ω is a paraboloid of revolution.

5 marks

- ✓(b) Suppose that a fluid with a density of 1000 kg/m³ flowing through a horizontal pipe at a speed of 5 m/s has a pressure of 100000 Pa. If this fluid then starts flowing through the pipe at a speed of 10 m/s, what is the new pressure that this fluid exerts?

3 marks



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Malaviya National Institute of Technology Jaipur

Department of Physics
B.Tech II Semester End Term Examination 2024-25

Course: Modern Physics

Course Code: 22PHT102

Date: April 30, 2025

Duration: 2 hour 30 minutes

Max. Marks: 50

Answer all the questions

1. (a) Write the statement of the Poynting's Theorem and prove it. (1+5)
(b) A current I is uniformly distributed over a copper wire of circular cross section, with radius a . Find the volume current density J . Find the total current in the wire if the current density in the wire is proportional to the distance from the axis, $J=ks$ (for some constant k). (2+2)
2. (a) Why is wave nature not observed in our daily life. (2)
(b) What is the difference between phase velocity and group velocity? (3)
(c) What is the de Broglie wavelength of electrons of energy 0.1 MeV using both relativistic and non-relativistic expression for λ ? What is the percentage error in using the non-relativistic expression for λ ? (5)
3. (a) Derive Schrödinger's time-dependent wave equation and separate it into space and time dependent parts. (3+2+2)
(b) Calculate the ground state energy of a linear harmonic oscillator using Heisenberg's uncertainty principle. (3)
4. (a) Solve the time independent Schrödinger's equation for a particle in a 1-D infinite potential box of width L . Find the normalized wave functions and determine the energy values. Find the expectation value of the position of the particle inside the box. (5+2)
(b) The energy of an electron constraint to move in a one dimensional potential box of length 4Å , is $9.664 \times 10^{-17}\text{J}$. Find the order of excited state and the momentum of electron in the state. (3)
5. (a) Determine the degree of degeneracy of the energy level $\frac{38\pi^2\hbar^2}{2mL^2}$ of a particle in a cubical potential box (3-D) of width L . (4)
(b) What do you mean by the orthogonality of the wave functions? Briefly explain the quantum tunneling. (2+4)

(Boltzmann constant: $1.38 \times 10^{-23} \text{ J K}^{-1}$; mass of electron: $9.1 \times 10^{-31} \text{ kg}$; $h=6.62 \times 10^{-34} \text{ Js}$)

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR

Department of Physics, End-term Examinations 2024, Even Semester

B. Tech: Modern Physics

Course Code: (22PHT102)

Time: 2.5 Hours

Max Marks: 50

Date: 3rd May 2024

Note: Answer all the questions.

- Q1. (a) Derive the wave equation for electromagnetic waves in free space. Prove that magnetic field vector $\vec{B}$ is perpendicular to both the propagation vector $\vec{k}$ and electric field vector $\vec{E}$. [6]
- (b) Write the Maxwell's equations in integral form. Using Faraday's law, determine $\vec{B}$, if $\vec{E} = xy^2e^{-t}\hat{z}$, and $\vec{B} = 0$ at $t = 0$. [4]
- Q2. (a) Derive an expression for the Poynting theorem and explain the physical significance of all of its terms. [6]
- (b) Derive the continuity equation in electrodynamics and discuss its physical significance. [4]
- Q3. (a) A parallelepiped sample of copper (resistivity $\rho = 1.6 \times 10^{-8} \Omega m$, length $L = 10 \text{ cm}$, width $W = 8 \text{ cm}$, height $H = 6 \text{ cm}$) is placed in a uniform magnetic field ($B_0(\hat{z})$) of 0.1 Tesla with its edges parallel to the axes. If an external emf ($V_0 = 10 \text{ V}$) is applied on the opposite faces along x -direction, then calculate the Hall voltage (V_H) developed in the sample (along $\hat{y}$) for the following orientations. ($R_H = -0.8 \times 10^{-10} m^3 A^{-1} s^{-1}$) [6]
- (i) $L(\hat{x}), W(\hat{y}), H(\hat{z})$; (ii) $W(\hat{x}), L(\hat{y}), H(\hat{z})$.
- (b) Explain the direct and indirect bandgap semiconductors with the help of E-k diagrams. [4]
- Q4. (a) State the Bloch theorem and explain the origin of band gap in solids with suitable equations and diagrams. [6]
- (b) There are 2.5×10^{28} free electrons per cubic meter of sodium. Calculate the Fermi energy and Fermi velocity. [4]
- Q5. (a) What is the lowest allowed energy and corresponding wave function for a particle in a 3D box with equal edge lengths? Calculate the degree of degeneracy for the first four energy states. [6]
- (b) A particle of mass m , which is confined to move in one dimension between 0 and L , is described by the wave function $\psi(x) = Ax(L-x)$. Normalize the wave function and find out the location of the maximum probability of finding the particle. Derive an expression for the average value of the position of the particle. [4]

(Boltzmann constant: $1.38 \times 10^{-23} \text{ J K}^{-1}$; mass of electron $9.1 \times 10^{-31} \text{ kg}$; $h = 6.62 \times 10^{-34} \text{ Js}$)

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Physics, End-term Examination 2019, Odd Semester

B. Tech: Physics (Theory) Course Code: (PHT-101) Time: 2 Hours 30 minutes Max Marks: 50

Answer any five questions

- Q1. (a) What is a wave function and what are the conditions for valid wave function?
 (b) An electron is trapped in a cubical box of side 1 Å. Find energy for ground state and first excited state.
 (c) In a radio transmitter, a pulse lasts for 1ms. Find the uncertainty in momentum and frequency of photons transmitted.

[4 + 3 + 3]

- Q.2 (a) Write Maxwell's equations in free space.

(b) Show that velocity of plane EM wave in free space is given by $C = 1/\sqrt{\mu_0 \epsilon_0}$, where symbols have their usual meaning.

(c) Suppose $\vec{v} = (xy)\hat{x} + (2yz)\hat{y} + (3zx)\hat{z}$. Check Stokes' theorem for the triangular shaded area as shown in the figure.



- Q.3 (a) What is the equation of continuity? Derive the relation.

[2 + 4 + 4]

(b) Calculate the power flow for a plane wave using the Poynting theorem.
 (c) Write the physical significance of the divergence, curl and gradient.

- Q.4 (a) A photon of frequency ν_0 scatters from an electron at rest and moves in a direction making an angle of 60° with the incident direction. If the frequency of scattered photon is half that of incident photon, calculate the frequency of the incident photon.

[4 + 3 + 3]

(b) What is Meissner effect?

(c) A superconducting tin has a critical temperature of 3.7 K at zero magnetic field and a critical field of 0.0306 Tesla at 0 K. Find the critical field at 2 K.

- Q5. (a) Define Fermi energy level and Explain Fermi Dirac distribution function.

[3 + 4 + 3]

(b) Derive the expression of density of states (DOS) of a metal in 3-D and show that it varies with energy as $\sqrt{E}$.

(c) What are direct band gap and indirect band gap type of semiconductors?

- Q6. (a) Explain the phenomenon of alpha decay.

[4 + 3 + 3]

(b) The electric field of a plane EM wave in vacuum is given by following expression:

$$E_x = \frac{1}{\sqrt{2}} \cos[4\pi \times 10^7 \left(t - \frac{z}{c}\right)]$$

Calculate wavelength, direction of polarization and energy flux. given that $E_y = E_z = 0$

(c) Briefly describe Type I and Type II superconductors.

[3 + 3 + 4]

(Boltzmann constant: $1.38 \times 10^{-23} \text{ J K}^{-1}$; mass of electron $9.1 \times 10^{-31} \text{ kg}$; $h=6.62 \times 10^{-34} \text{ Js}$)

Malaviya National Institute of Technology Jaipur
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Department of Physics

End Term Examination (Odd Semester/ 2018-19)

Class: B.Tech (Physics)

Subject: Physics (PHT-101)

Max. Marks: 50

Time: 2 Hrs.

Note: All questions are to be answered. Internal choice is given in Q. No. 4.

Q. No. 6 is of 10 Marks.

- Q.1 (a) Distinguish between a free particle and a particle in a box in terms of eigen functions. (3)
- (b) A particle of mass 1.67×10^{-27} kg is confined to the second excited state ($n=3$) in a one dimensional potential well of infinite height and width $L=0.1$ nm. Calculate its energy, momentum and the probability of finding the particle between 0 and $L/3$. (5)
- Q.2 (a) At what scattering angle will the 100 keV X-rays incident on a target will get Compton scattered with an energy of 90 keV. (3)
- (b) Define Einstein coefficients and obtain the relation between them. (5)
- Q.3 (a) Why nanomaterials have different properties than that of their bulk counterpart. (3)
- (b) Given that the effective masses of electrons and holes in Si are approximately $1.08 m_e$ and $0.60 m_e$ respectively, and the electron and hole drift mobilities at room temperature are 1350 and $450 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, respectively, calculate the intrinsic concentration and intrinsic resistivity of Si. (5)
- Q.4 (a) Derive an expression of the continuity equation and explain the conservation of charge. (4)
- (b) Calculate the mean value of intensity of the magnetic field in air at a distance of 1 m from a radiating source of power of 100 W . Given ($\mu_0=4\pi \times 10^{-7} \text{ H/m}$, $\epsilon_0=8.85 \times 10^{-12} \text{ F/m}$)
- OR
- (b) Find the magnetic field B of electromagnetic waves in free space if the components of the electric fields are $E_x = E_y = 0$ and $E_z = E_0 \cos kx \sin \omega t$.
- Q.5 (a) Explain London penetration depth (λ_L) for a superconductor. What is the order of λ_L generally? (4)
- (b) Give a brief account of important characteristics of a superconductor. (4)
- Q.6 (a) Explain Hall Effect in a semiconductor. Derive the expression for Hall Coefficient for an intrinsic semiconductor, and thereby show that the sign of the Hall Coefficient is determined by the carriers with greater mobility. (5)
- (b) Give postulates of special theory of relativity. Based on these postulates, derive Lorentz Transformation equations of space and time. (5)

Malaviya National Institute of Technology Jaipur

Department of Physics

End Term Examination (Even Semester / 2018 - 2019)

Class: B Tech I (Physics)

Subject: Physics (PHT-101)

Max. Marks: 50

Time: 2 hours

Note: Attempt any five questions.

Q.1

6+4

- (a) Write the Maxwell's equations in free space. Show that velocity of plane EM wave in free space is given by $C = 1/\sqrt{\mu_0\epsilon_0}$, where symbols have their usual meaning.
(b) Prove the divergence theorem using the function $F = y^2 \hat{i} + (2xy + z^2) \hat{j} + (2yz) \hat{k}$ and the unit cube situated at origin.

Q.2

4+2+2+2

- (a) Solve the Schrodinger equation for a particle trapped in 1D box and show that particle takes discrete energies.
(b) A beam of γ rays of energy 450 keV is incident on a foil of Al. Calculate the wavelength of scattered γ rays at 90° and energy of scattered photon.
(c) An electron is trapped in a cubical box of side 1 Å. Find energy for ground state and first excited state.
(d) Explain quantum tunneling through a rectangular potential barrier.

Q.3:

6+4

- (a) Derive expression for carrier concentration in an intrinsic semiconductor (for electrons) and prove that Fermi energy level lies at the mid of the band gap.
(b) Calculate the intrinsic concentration of charge carriers at 300 K. Given that $m_e^* = 0.12m_0$, $m_p^* = 0.28m_0$ and the band gap for Ge is $E_g = 0.67\text{eV}$.

Q.4

7+3

- (a) What is Hall Effect? Explain the charge accumulation process in an intrinsic semiconductor with a neat diagram. How the effect is used in deciding the sign and density of charge carrier in a conducting material.
(b) The resistivity of a doped Si sample is $8.9 \times 10^{-3} \Omega\text{m}$. The Hall coefficient was measured to be $3.6 \times 10^{-4} \text{m}^3/\text{C}$. Assuming single carrier conduction, find the mobility and density of charge carriers.

Q.5

6+4

- (a) Describe the laser action in Ruby laser with energy level diagram. An optical fibre has refractive index of core 1.5 and the relative refractive index difference of core cladding is 0.01. Find the numerical aperture and maximum angle of acceptance.
(b) Write short note on any two:
(i) Meissner effect (ii) Superconductivity (iii) Nanostructured materials

Q.6:

7+3

- (a) Using the principle of special theory of relativity derive mass energy relation and illustrate it with example.
(b) A laboratory observer shows that the length of 2m is crossed by a beam of radioactive particles in $1 \times 10^{-8} \text{s}$ and, in this process, half of the particles disintegrate. Calculate the proper half-life. Also, calculate the transverse length as seen by an observer fixed relative to the particle.

Malaviya National Institute of Technology Jaipur

Department of Physics

B. Tech Ist Year Physics (Theory) End Term Exam (Autumn Semester) 2013-14

Duration:- 2 Hrs

M.M: 50

Part A- Attempt any four questions

- 1 (a) Explain the requirements that are imposed on a physically acceptable wave function when the wave function is said to be normalized. (2+1)
- (b) Derive the expression for Schrodinger time independent wave equation. (4)
- 2 Give complete theory of Compton effect & deduce the expression for the change in wave length of the scattered photon. (7)
- 3 (a) Obtain an expression for the carrier density of an intrinsic semiconductor. (4)
- (b) Indicate on an energy level diagram the conduction & valence bands, fermi level & acceptor states for an intrinsic, n- type and p-type semiconductor. (3)
- 4 (a) Explain what is meant by nanostructured materials. Give their important characteristics & applications. (3)
- (b) State and explain the basic laws of electromagnetism in their differential form. (4)
- 5 (a) Explain spatial & temporal coherence. Deduce the relation between coherence length & line width $\Delta\lambda$. (3)
- (b) What is uncertainty principle? Apply it to prove the nonexistence of electron in the nucleus. (4)

Part B- Attempt any four questions

- 1 (a) Calculate the mass m & speed v of an electron having kinetic energy 1.5 MeV. Given, the rest mass of electron, $m_e = 9.1 \times 10^{-31}$ Kg. (3½)
- (b) Name two applications of Einstein mass energy equivalence. (2)
- 2 (a) An Electric field of 100V/m is applied to a sample of n-type semiconductor whose Hall coefficient is $-0.0125 \text{ m}^3/\text{C}$. Determine the current density in the sample assuming $m_e = 0.36 m^*/V/\text{sec}$. (3)
- (b) A ruby laser emits light of 693.96 nm wavelength. If one of a mole of Cr^{3+} ions are involved in population inversion process in a pulse, calculate the pulse energy in eV. (2½)
- 3 Calculate the current density for 1 mm diameter of wire of lead at 4.2 K. Given: $T_c = 7.18 \text{ K}$ & $H_c = 6.5 \times 10^4 \text{ A/m}$. (5½)
- 4 (a) Calculate the minimum energy of a neutron confined to the nucleus, considering the nucleus as a potential box with a size of 10^{-14} m . ($h = 6.63 \times 10^{-34} \text{ J-s}$, Neutron Mass $= 1.67 \times 10^{-27} \text{ kg}$). (4½)
- (b) What is the approximate energy required to detach the fifth valence electron from the Arsenic impurity atom surrounded by Ge Atom. (1)
- 5 A particle is moving in one dimensional potential box (of infinite height) of width 25 Å . Calculate the probability of finding the particle within an interval of 5 Å at the centre of the box when it is in its state of least energy. (5½)